VELOOP REWARDS - Simulated Data Profiling

Data Analyst Internship, Task 01 | Deliverable 12 | Work Package J

SIMULATED DATA - CREATED FOR ANALYTICAL TESTING. The profiled file is fictional. Every number below describes the supplied simulated dataset only and is not an actual VELOOP statistic. Personal-looking fields (name, email, age, gender, state, city) exist in the file; their values are not reproduced here.

1. Summary

- **The file contains the Users entity only**: a clean sheet (1,000 rows) and an audit sheet (1,008 rows), plus a 25-field dictionary and a summary sheet. The Task Activity, Rewards, Wallet, Withdrawals, Referrals and Feature Events sheets from the brief are not in it (A046).
- **KPI test:** of the 54 KPIs, 2 can be calculated in full, 5 partly (from proxies) and 47 cannot, because the data they need was not supplied.
- **Relationship tests:** inside Users the keys behave (user_id and referral_code are unique in the clean sheet; no orphans between the two sheets). Tests against the six missing sheets cannot be run, and referrals cannot be linked to a referrer.
- **The 'clean' sheet is not free of rule violations** (negative derived days, logins after the period end, sessions without a login, and others). The Data Quality Audit lists them.
- **The dataset's own summary is accurate:** all 8 summary metrics match a recomputation.

2. What was expected and what was supplied

Dataset in the brief	Approx. records	What the file contains	Result
Users	1,000	Users_1000_Clean: 1,000 rows	Found
Task Activity	6,000	Not in the file	Missing
Rewards	5,000	Not in the file	Missing
Wallet	3,500	Not in the file	Missing
Withdrawals	1,200	Not in the file	Missing
Referrals	900	Not in the file	Missing
Feature Events	12,000	Not in the file	Missing
Messy Data	500+	Users_Messy_Audit: 1,008 rows (all users, with injected problems)	Found, in a different form

Sheets in the supplied file

Sheet	Rows	Columns	Role
Users_1000_Clean	1000	27	Data
Users_Messy_Audit	1008	28	Data (with injected problems)
Data_Dictionary	25	5	Reference
Synthetic_Summary	8	2	Reference
README_Data_Status	7	2	Reference

3. J1 Profile of each dataset

Measure	Users_1000_Clean	Users_Messy_Audit
Records	1000	1008
Columns	27	28
Registration range	01 Apr 2026 to 15 Sep 2026	01 Apr 2026 to 15 Sep 2026
Last-login range	15 Apr 2026 to 17 Sep 2026	15 Apr 2026 to 17 Sep 2026
Unique user_id	1000	1000
Duplicate rows	0	8
Duplicate user_id values	0	8
Missing cells	953	966

Numeric ranges

Dataset	Field	Non-null	Min	P25	Median	P75	Max	Mean	Zeros	Negatives
Users_1000_Clean	age	1000	18	23	28	33	38	28.0	0	0
Users_1000_Clean	lifetime_sessions	1000	0	5	9	18	81	12.9	30	0
Users_1000_Clean	active_days	1000	0	3	6	11	32	7.7	121	0
Users_1000_Clean	xp_balance	1000	5	109	191.5	305.8	1156	231.9	0	0
Users_1000_Clean	days_since_registration	1000	-0.8	39.9	79.4	124.6	166.9	81.3	1	4
Users_Messy_Audit	age	1007	18	23	28	33	38	28.1	0	0
Users_Messy_Audit	lifetime_sessions	1008	0	5	9	18	81	12.8	30	0
Users_Messy_Audit	active_days	1008	0	3	6	11	32	7.7	122	0
Users_Messy_Audit	xp_balance	1008	5	108.8	191	305.8	1156	231.8	0	0
Users_Messy_Audit	days_since_registration	1008	-0.8	40.0	79.4	124.5	166.9	81.4	1	4

Registrations per month (simulated; September is partial)

Month	Clean sheet	Audit sheet (distinct users)
2026-04	174	174
2026-05	179	178
2026-06	166	166
2026-07	198	198
2026-08	192	191
2026-09	91	91

Status and category distributions (clean sheet; audit sheet distinct users)

Account status

Value	Clean count	Clean share	Audit sheet (distinct users)
Active	568	56.8%	568
Inactive	283	28.3%	283
Suspended	77	7.7%	77
Pending Verification	72	7.2%	72

Email verification

Value	Clean count	Clean share	Audit sheet (distinct users)
Verified	855	85.5%	855
Pending	145	14.5%	144

Onboarding status

Value	Clean count	Clean share	Audit sheet (distinct users)
Completed	722	72.2%	721
Started	146	14.6%	146
Not Started	132	13.2%	132

Acquisition source

Value	Clean count	Clean share	Audit sheet (distinct users)
Organic Search	244	24.4%	244
Direct	217	21.7%	216
Social Media	160	16.0%	160
Referral	119	11.9%	119
WhatsApp	103	10.3%	103
Other	102	10.2%	102
Telegram	55	5.5%	55

Account created via

Value	Clean count	Clean share	Audit sheet (distinct users)
Web Registration	661	66.1%	661
Google Sign-In	339	33.9%	339

Primary device

Value	Clean count	Clean share	Audit sheet (distinct users)
Android	659	65.9%	658
Windows	155	15.5%	155
iOS	140	14.0%	140

Value	Clean count	Clean share	Audit sheet (distinct users)
macOS	46	4.6%	46

Consent status

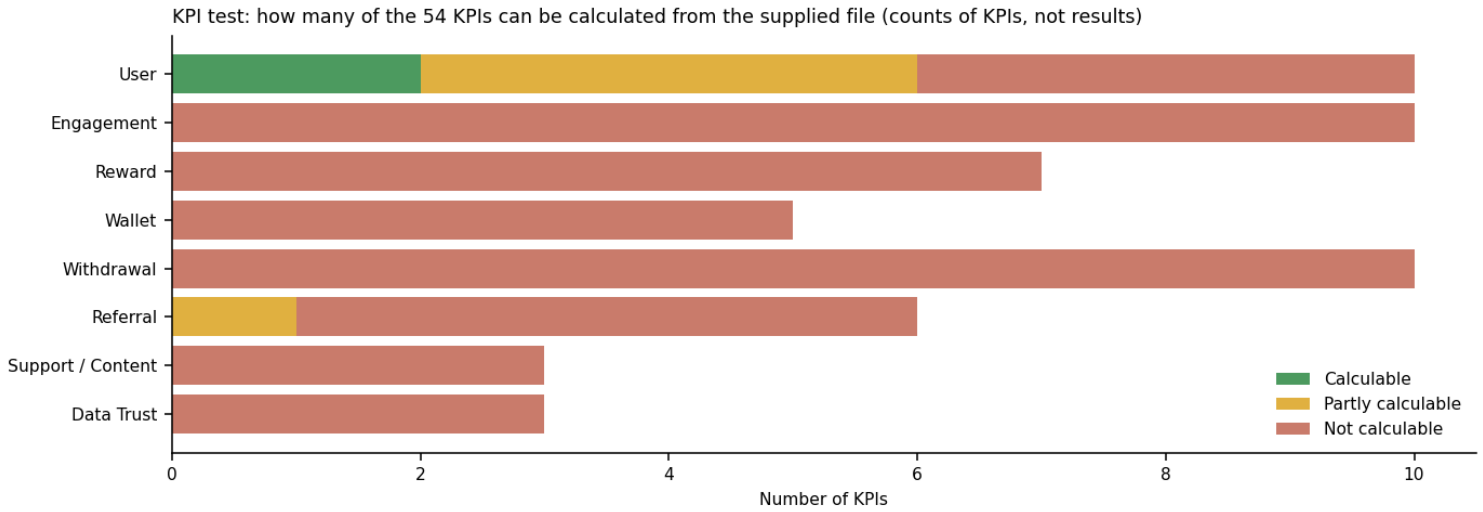
Value	Clean count	Clean share	Audit sheet (distinct users)
Accepted	923	92.3%	923
Pending	77	7.7%	77

4. J2 Relationship tests

Test	Relationship tested	Result	Detail
J2-01	Users_1000_Clean.user_id is unique	Pass	0 duplicate user_id values in 1,000 rows
J2-02	Users_Messy_Audit.user_id is unique	Fail	8 user_id values repeat (8 ids appear twice)
J2-03	Every Users_Messy_Audit user_id exists in Users_1000_Clean (orphan check)	Pass	0 orphan rows; all 1000 distinct messy ids are in the clean sheet
J2-04	Every Users_1000_Clean user_id exists in Users_Messy_Audit	Pass	0 clean ids missing from the messy sheet
J2-05	referral_code is unique per user in Users_1000_Clean	Pass	0 duplicate codes
J2-06	email is unique per user in Users_1000_Clean	Pass	0 duplicate values (values not shown)
J2-07	Each city belongs to one state (Users_1000_Clean)	Pass	29 cities; 0 map to more than one state
J2-08	referral_source_detail is filled exactly for users acquired by Referral	Pass	119 Referral users, 119 with a detail value
J2-09	A referred user can be linked to the user who referred them	Fail	referral_source_detail holds only 'Friend/Family'. There is no referrer user_id or referral_code_used, so a referral link cannot be built
J2-10	Users.user_id connects to Task Activity.user_id	Not testable	The Task Activity sheet was not supplied
J2-11	Users.user_id connects to Rewards.user_id	Not testable	The Rewards sheet was not supplied
J2-12	Users.user_id connects to Wallet.user_id	Not testable	The Wallet sheet was not supplied
J2-13	Users.user_id connects to Withdrawals.user_id	Not testable	The Withdrawals sheet was not supplied
J2-14	Users.user_id connects to Referrals.user_id	Not testable	The Referrals sheet was not supplied
J2-15	Users.user_id connects to Feature Events.user_id	Not testable	The Feature Events sheet was not supplied

5. J3 KPI test: can each KPI be calculated from the supplied structure?

A KPI is Calculable if every required field exists in the file, Partly calculable if only a proxy exists, and Not calculable otherwise. Simulated test results only prove calculability.



KPI test detail

ID	KPI	Verdict	Why	Additional data required	Simulated test result
U01	Registrations	Calculable	All required fields exist in the supplied data.	-	Registrations per month: 2026-04: 174, 2026-05: 179, 2026-06: 166, 2026-07: 198, 2026-08: 192, 2026-09: 91
U02	Daily Active Users (DAU)	Not calculable	Needs data that is not in the supplied file: Event-level activity with timestamps (qualifying events per user per day).	Event-level activity with timestamps (qualifying events per user per day)	
U03	Weekly Active Users (WAU)	Partly calculable	Snapshot count from is_active_7d (users whose last login is within 7 days of the reference time). It measures logins, not qualifying activity, and gives one value, not a time series.	Event-level activity to compute a weekly series	78 users flagged
U04	Monthly Active Users (MAU)	Partly calculable	Snapshot count from is_active_30d (login within 30 days of the reference time). Logins, not qualifying activity; one value, not a trend.	Event-level activity to compute a monthly series	325 users flagged
U05	Returning Users	Not calculable	Needs data that is not in the supplied file: Event-level activity plus first_activity_at.	Event-level activity plus first_activity_at	
U06	Inactive Users	Partly calculable	Snapshot using last_login_datetime and the 30-day flag; 'Inactive' as an account_status is not the same thing. The window is not agreed (A024).	last_activity_at from events and an agreed inactivity window	675 users with no login in 30 days (includes 72 who never logged in)
U07	Onboarding Completion Rate	Partly calculable	Snapshot share with onboarding_status = Completed. There are no start or completion times and no steps, so it is not a funnel.	onboarding_started_at, onboarding_completed_at and onboarding_step	72.2% of users have onboarding Completed
U08	Cohort Retention (Day N)	Not calculable	Needs data that is not in the supplied file: Daily activity per user after registration.	Daily activity per user after registration	
U09	Account Status Mix	Calculable	All required fields exist in the supplied data.	-	Account status counts: Active: 568, Inactive: 283, Suspended: 77, Pending Verification: 72
U10	Daily Visitors	Not calculable	Needs data that is not in the supplied file: Login events with timestamps (one per login).	Login events with timestamps (one per login)	
E01	Feature Usage	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E02	Feature Adoption Rate	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E03	Task Starts	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E04	Task Completions	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E05	Task Completion Rate	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E06	Task Failure Mix	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E07	Repeat Usage Rate	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E08	Mission Completion Rate	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E09	Streak Length	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
E10	Task Open Rate	Not calculable	Needs data that is not in the supplied file: Task Activity and Feature Events tables.	Task Activity and Feature Events tables	
R01	Reward Transactions	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	

ID	KPI	Verdict	Why	Additional data required	Simulated test result
R02	Reward Credits	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
R03	Reward Reversals	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
R04	Reward Reversal Rate	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
R05	Reward Activity by Feature	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
R06	Pending Rewards	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
R07	Net Reward Amount	Not calculable	Needs data that is not in the supplied file: Rewards table (reward records with amount, unit, status).	Rewards table (reward records with amount, unit, status)	
W01	Wallet Transactions	Not calculable	Needs data that is not in the supplied file: Wallet_Transactions table.	Wallet_Transactions table	
W02	Credit Volume	Not calculable	Needs data that is not in the supplied file: Wallet_Transactions table.	Wallet_Transactions table	
W03	Redemption Activity	Not calculable	Needs data that is not in the supplied file: Wallet_Transactions table.	Wallet_Transactions table	
W04	Wallet Reversals	Not calculable	Needs data that is not in the supplied file: Wallet_Transactions table.	Wallet_Transactions table	
W05	Exchange Activity	Not calculable	Needs data that is not in the supplied file: Wallet_Transactions table.	Wallet_Transactions table	
D01	Withdrawal Requests	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D02	Completed Withdrawals	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D03	Pending Withdrawals	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D04	Rejected Withdrawals	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D05	Reversed Withdrawals	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D06	Processing Time	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D07	Withdrawal Completion Rate	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D08	Pending Queue Age	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D09	Withdrawal Volume	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
D10	Withdrawal Method Distribution	Not calculable	Needs data that is not in the supplied file: Withdrawals table with state timestamps.	Withdrawals table with state timestamps	
F01	Referrals Created	Not calculable	Needs data that is not in the supplied file: Referrals table.	Referrals table	
F02	Referral Registrations	Partly calculable	Count of users whose acquisition_source is Referral. There is no referral_id, referrer or timestamp, so referral registrations cannot be linked to referrals.	Referrals table: referral_id, referring and referred user, timestamps	119 users acquired by referral
F03	Referral Activations	Not calculable	Needs data that is not in the supplied file: Referrals table.	Referrals table	
F04	Referral Conversions	Not calculable	Needs data that is not in the supplied file: Referrals table.	Referrals table	
F05	Referral Activation Rate	Not calculable	Needs data that is not in the supplied file: Referrals table.	Referrals table	
F06	Referral Conversion Rate	Not calculable	Needs data that is not in the supplied file: Referrals table.	Referrals table	

ID	KPI	Verdict	Why	Additional data required	Simulated test result
S01	Support Requests	Not calculable	Needs data that is not in the supplied file: Support_Tickets table and Help Center events.	Support_Tickets table and Help Center events	
S02	Issue Category Mix	Not calculable	Needs data that is not in the supplied file: Support_Tickets table and Help Center events.	Support_Tickets table and Help Center events	
S03	Help-Center Engagement	Not calculable	Needs data that is not in the supplied file: Support_Tickets table and Help Center events.	Support_Tickets table and Help Center events	
T01	Wallet Reconciliation Exceptions	Not calculable	Needs data that is not in the supplied file: Wallet, provider callback or event receive-time data.	Wallet, provider callback or event receive-time data	
T02	Provider Callback Reliability	Not calculable	Needs data that is not in the supplied file: Wallet, provider callback or event receive-time data.	Wallet, provider callback or event receive-time data	
T03	Event Delivery Delay	Not calculable	Needs data that is not in the supplied file: Wallet, provider callback or event receive-time data.	Wallet, provider callback or event receive-time data	

6. Schema alignment: supplied dataset vs the proposed data dictionary

Dataset field	Proposed field	Match	Dataset values	Recommendation
user_id	Users.user_id	Same meaning	String; format VLUSR + 6 digits; unique in the clean sheet	Matches our requirement for a stable, opaque ID. Keep.
full_name	(not in dictionary)	Not in our dictionary	Personal-looking	Excluded by design (Work Package B). Do not carry into analytics.
email	(not in dictionary)	Not in our dictionary	Personal-looking	Excluded by design. Use email_verification_status instead where verification matters.
age	(not in dictionary)	Not in our dictionary	Integer 18 to 38	Not needed for any KPI. Add only if a need is shown (A048).
gender	(not in dictionary)	Not in our dictionary	Male, Female, Prefer not to say	Not needed for any KPI (A048).
state	(not in dictionary)	Not in our dictionary	17 states	We proposed country level only. State is finer than needed (A048).
city	(not in dictionary)	Not in our dictionary	29 cities, each in one state	Finer than needed (A048).
registration_datetime	Users.registered_at	Renamed	Datetime, no timezone	Rename; store in UTC with a timezone (A012).
acquisition_source	Users.registration_source	Renamed; values differ	Organic Search, Direct, Social Media, Referral, WhatsApp, Other, Telegram	Adopt these as the first approved list, then confirm (A022).
referral_source_detail	(not in dictionary)	Not in our dictionary	Friend/Family, only for Referral users	No referrer link. We need referral_code_used and a Referrals table to measure conversion.
primary_device	(not in dictionary)	New field	Android, iOS, Windows, macOS	Useful device attribute; our platform field (android, ios, web) is different. Keep both.
platform	Users.platform	Same name; different values	Always 'Web'	Our values are android, ios, web. Clarify whether this is the channel or the device.
account_status	Users.account_status	Same name; values differ	Active, Inactive, Suspended, Pending Verification	Ours: active, restricted, closed. Add 'pending_verification'. 'Inactive' clashes with the derived inactive concept (A047).
email_verification_status	(not in dictionary)	New field	Verified, Pending	Useful for onboarding analysis. Propose adding to Users.
onboarding_status	Users.onboarding_status	Same name; values differ	Completed, Started, Not Started	Ours: not_started, in_progress, completed, skipped. Align names (A019).
last_login_datetime	Users.last_login_at	Renamed	Datetime; empty for Pending Verification	Rename. A latest-login field exists but not event-level logins.
lifetime_sessions	(not in dictionary)	New field	Integer 0 to 81	A counter. Better derived from events than stored.
active_days	(not in dictionary)	New field	Integer 0 to 32	A counter. Better derived from events than stored.
xp_balance	(not in dictionary)	New field	Integer 5 to 1,156	A balance, not transactions. XP activity needs a transactions table.
referral_code	(not in dictionary)	New field	VL + 6 digits; unique	The user's own code. We also need referral_code_used to link a referred user.
days_since_registration	(derived)	Derived field stored	Decimal; 4 negative values	Do not store; compute in the analytics layer.
is_active_7d	(derived)	Derived field stored	Yes or No	Do not store; derive lifecycle and activity from events.
is_active_30d	(derived)	Derived field stored	Yes or No	Do not store; derive from events.
consent_status	(not in dictionary)	New field	Accepted, Pending	Important for governance; consider adding (A048).
marketing_opt_in	(not in dictionary)	New field	Yes, No	Consider adding, with a consent consistency rule (DQ042).
account_created_via	Users.registration_method	Renamed	Web Registration, Google Sign-In	Rename; category only, which matches our design.
country	Users.country_code	Renamed	Always India	Rename to an ISO country code (IN).
data_environment	(not in dictionary)	New field (messy sheet only)	'SIMULATED — NOT ACTUAL VELOOP DATA' on every row	A good per-row label. Recommend the clean sheet carries it too (DQ039).

Proposed fields or tables not found in the file

Proposed field or table	Entity	Status
app_version	Users	Not supplied
language_code	Users	Not supplied
referral_code_used	Users	Not supplied; needed to link a referred user to a referral
campaign_id	Users	Not supplied

Proposed field or table	Entity	Status
onboarding_started_at / onboarding_completed_at	Users	Not supplied; needed for a funnel
first_activity_at / first_reward_credited_at / last_activity_at	Users	Not supplied; derived from events
status_changed_at / account_closed_at	Users	Not supplied; no status history
Task, reward, wallet, withdrawal, referral, event and support tables	All other entities	Not supplied

7. Dataset summary check

Metric	Value stated	Recomputed	Match
Users	1000	1000	Match
Active 7d	78	78	Match
Active 30d	325	325	Match
Verified email	855	855	Match
Onboarding completed	722	722	Match
Google Sign-In	339	339	Match
Referral acquisition	119	119	Match
Suspended	77	77	Match

8. Findings

ID	Finding	Evidence	Implication
F01	The file holds only the Users entity: a clean sheet (1,000 rows) and an audit sheet (1,008 rows). The Task Activity, Rewards, Wallet, Withdrawals, Referrals and Feature Events sheets described in the brief are not in it.	6 of the 8 expected datasets missing	Relationship tests beyond Users and 47 of 54 KPI tests cannot be run (A046).
F02	KPI test: 2 KPIs are fully calculable (Registrations, Account Status Mix), 5 are partly calculable from proxies, and 47 are not calculable.	2 calculable, 5 partly, 47 not	The structure supplied supports user counting only; earning, reward, wallet, withdrawal and engagement analytics need their own data.
F03	The dataset's own Synthetic_Summary matches a recomputation from the clean sheet for all 8 metrics.	8 of 8 match	The summary can be trusted as a description of the clean sheet.
F04	Stored derived fields (days_since_registration, is_active_7d, is_active_30d) were computed with a reference time of 15 Sep 2026 00:00, which the file does not state. The 7-day and 30-day counts only reproduce with that time.	3 derived fields	Derived values depend on an undocumented reference time (A049). Compute them in the analytics layer instead.
F05	The 'clean' sheet still breaks rules in the dataset's own dictionary or in common sense: 4 negative days_since_registration values, 4 logins after the stated period end, 72 users with sessions but no login, 30 users with XP but no session, many users with more active days than days available, and 34 users who opted in to marketing while consent is Pending.	See the Data Quality Audit	'Clean' here means free of injected problems, not free of rule violations.
F06	account_status and activity disagree: 25 Suspended and 95 Inactive accounts are flagged active in the last 30 days, and 363 Active accounts are not.	120 contradictory accounts	Supports deriving activity from events and keeping account state separate (Work Package B lifecycle model).
F07	Referral acquisition has no link to a referrer. referral_source_detail is always 'Friend/Family' and no referral_code_used exists.	119 referral users unlinked	Referral activation and conversion cannot be measured until a Referrals table with referrer and referred ids exists.
F08	platform is 'Web' for every user and country is 'India' for every user, while primary_device varies (Android, iOS, Windows, macOS). Neither platform nor country is listed in the dataset's own Data_Dictionary (25 fields documented, 27 columns present).	2 constant, undocumented fields	platform carries no information here. Our platform values (android, ios, web) need a clear meaning.
F09	The file holds personal-looking fields (name, email, age, gender, state, city) that our proposed data dictionary deliberately excludes. This profile does not show their values.	6 fields	Data minimisation: share only a sanitised version with analysts and interns (Work Package P).
F10	Datetime values carry no timezone.	1,928 values	A day boundary cannot be applied safely until the timezone is stated (A012).
F11	The audit sheet is the clean sheet plus 8 repeated rows and 12 changed cells (8 blanked values, 1 missing, 1 malformed and 2 altered registration times).	8 + 12 injected problems	All messy user_ids exist in the clean sheet, so the injected problems can be traced exactly (see the Data Quality Audit).
F12	The dataset adds fields that our dictionary does not have: email_verification_status, consent_status, marketing_opt_in, primary_device, lifetime_sessions, active_days, xp_balance and referral_code.	8 new fields	Consider adding verification and consent fields; treat the counters as derived (see Schema_Alignment).

9. What is needed next

- Supply the six missing datasets (task activity, rewards, wallet, withdrawals, referrals, feature events), or confirm they do not exist yet (A046). Then re-run the relationship tests and the KPI test.
- State the reference time and timezone behind the derived fields (A049, A012).
- Decide which of the new dataset fields (email verification, consent, device) should join the proposed dictionary, and which personal-looking fields should not (A048).